1.Program to print the area of rectangle by creating a class named "Area" taking the values of ite length and breadth as parameters of its constructor and having a method named "returnedArea" which returns the area of the rectangle. Length and breadth of rectangle are entered through keyboard.

```
class Area:
    def __init__(self, length, breadth):
        self.length = length
        self.breadth = breadth

    def returnedArea(self):
        return self.length * self.breadth

length = float(input("Enter length: "))
breadth = float(input("Enter breadth: "))


rect = Area(length, breadth)


print("Area of rectangle:", rect.returnedArea())
```

```
Enter length: 5
Enter breadth: 8
Area of rectangle: 40.0
```

```
from abc import ABC
class Test(ABC):
  def m1(self):
    pass
  def m2(self):
    pass
class MyPython(Test):
  def m1(self):
    print("m1 implementation")
  def m2(self):
    print("m2 implementation")

m=MyPython()
m.m1()
m.m2()
```

```
m1 implementation
m2 implementation
```

```
from abc import ABC
class Test(ABC):
  def m1(self):
    pass
  def m2(self):
    pass
class MyPython(Test):
  def m1(self):
    print("m1 implementation")
  def m2(self):
    pass
class child(MyPython):
  def m2(self):
      print("m2 implementation")

m=child()
m.m1()
m.m2()
```

```
m1 implementation
m2 implementation
```

```
from abc import ABC
class Test(ABC):
  def __init__(self):
    print("Constructor is called...")
  def m1(self):
    pass
```

```
class MyPython(Test):
  def m1(self):
    print("m2 Method implementation")
m=MyPython()
m.m1()
```

```
Constructor is called...
m2 Method implementation
```

```
class Test1:
    def __init__(self):
        self.a = 101
        self._b = 201
        self.__c = 301  # Private to Test1 (mangled to _Test1__c)

    def show1(self):
        print("a=", self.a)
        print("b=", self._b)
        print("c=", self.__c)


class Test2(Test1):
    def show2(self):
        print("a=", self.a)
        print("b=", self._b)
        print("c=", self._Test1__c)  # Accessing mangled private attribute


class Test3(Test2):
    def show3(self):
        print("a=", self.a)
        print("b=", self._b)
        print("c=", self._Test1__c)


t1 = Test1()
t1.show1()

t2 = Test2()
t2.show2()

t3 = Test3()
t3.show3()
```

```
a= 101
b= 201
c= 301
a= 101
b= 201
c= 301
a= 101
b= 201
c= 301
```

2.Second largest element.

```
mylist=[7,3,9,2,8]
mylist.sort()
print(mylist[-2])
```

```
8
```

```
list1=[0,1,0,3,12]
for i in list1:#
```

```
[1, 3, 12, 0, 0]
```

```
def f(i,values=[]):
  values.append(i)
  print(values)
f(1)
f(2)
f(3)
```

```
[1]
[1, 2]
[1, 2, 3]
```

```
fruits_list1 = ['Apple', 'Berry', 'Cherry', 'Papaya']
fruits_list2 = fruits_list1
fruits_list3 = fruits_list1[:]

fruits_list2[0] = 'Guava'
fruits_list3[1] = 'Kiwi'

sum = 0
for ls in (fruits_list1, fruits_list2, fruits_list3):  # Fixed typo here
    if ls[0] == 'Guava':
        sum += 1
    if ls[1] == 'Kiwi':
        sum += 20

print(sum)
```

```
22
```

```
data=[[1,2],[3,4]],[[5,6],[7,8]]
def fun(m):
  v=m[0][0]
  for row in m:
    for element in row:
      if v < element:v=element
  return v
print(fun(data[0]))
```

```
4
```

```
a={(1,2):1,(2,3):2,(4,5):3}
print(a[4,5])
```

```
3
```

```
a={'a':1,'b':2,'c':3}    #only one key can call at a time.

print(a['a','b'])
```

```
---------------------------------------------------------------------------
KeyError                                  Traceback (most recent call last)
/tmp/ipykernel_3197/1673002352.py in <cell line: 0>()
      1 a={'a':1,'b':2,'c':3}    #only one key can call at a time.
      2
----> 3 print(a['a','b'])

KeyError: ('a', 'b')
```

```
fruit={}
def addone(index):
  if index in fruit:
    fruit[index]+=1
  else:
    fruit [index]=1
addone ('Apple')
addone('Banana')
addone('apple')
print(len(fruit))
```

```
3
```

```
arr={}
arr[1]=1
arr['1']=2
arr[1]+=1
print(arr)
sum=0
for K in arr:
  sum+=arr[K]
print(sum)
```

```
{1: 2, '1': 2}
4
```

```
my_dict={}
my_dict[1]=1
my_dict['1']=2
my_dict[1.0]=4
print(my_dict)
sum=0
for k in my_dict:
  sum+=my_dict[k]
print(sum)
```

```
{1: 4, '1': 2}
6
```

```
my_dict={}
my_dict[(1,2,4)]=8
my_dict[(4,2,1)]=10
my_dict[(1,2)]=12
print(my_dict)
sum=0
for k in my_dict:
  sum+=my_dict[k]
print(sum)
```

```
{(1, 2, 4): 8, (4, 2, 1): 10, (1, 2): 12}
30
```

```
box = {}
jars = {}
crates = {}

box['biscuit'] = 1
box['cake'] = 3
jars["jam"] = 4

crates['box'] = box
crates['jars'] = jars

print(len(crates['box']))
```

```
2
```

```
dict={'c':97,'a':96,'b':98}
for _ in sorted(dict):
  print(dict[_])
```

```
96
98
97
```

```
rec={"Name":"Python", "Age":"20"}
r=rec.copy()
print(id(r)==id(rec))
```

```
False
```

```
rec={"Nmae":"Python","Age":"20","Addr":"NJ","Country":"USA"}
id1=id(rec)
del rec
rec={"Name":"Python","Age":"20","Addr":"NJ","Country":"USA"}
id2=id(rec)
print(id1==id1)
```

```
True
```

```
rollno=[3,5,7,11,4,5,2]
for x in rollno:
  if x==2 or x==4 or x==6 or x==8 or x==10:
    print("Which even no is found",x)
```

```
Which even no is found 4
Which even no is found 2
```

```
arr=[5,4,2,1]
for i in arr:
  print(i,end=" ")
```

```
5 4 2 1
```

```
for i in range (5,0,-1):
  if i==3:
    continue
  print(i,end=" ")
```

```
5 4 2 1
```

```
for i,j in zip (range(1,6),range(5,0,-1)):
  if i==3 and j==3:
    continue
  print(i," ",j)
```

```
1   5
2   4
4   2
5   1
```

FILE HANDLING

```
f=open("myfile.txt",'w')
f.write("Ashish\n")
f.write("Amar\n")
f.write("Akash\n")
print("Write operation done successfully")
f.close()
```

```
Write operation done successfully
```

```
f=open("myfile.txt",'r')
data=f.read()
print(data)
f.close()
```

```
Ashish
Amar
Akash
```

```
f=open("myfile.txt",'r')
data=f.read(12)
print(data)
f.close()
```

```
Ashish
Amar
```

```
f=open("myfile.txt",'r')
l1=f.readlines()
print(l1)
for x in l1:
  print(x,end=" ")
f.close()
```

```
['Ashish\n', 'Amar\n', 'Akash\n']
Ashish
 Amar
 Akash
```

```
with open("myfile.txt",'w') as f:
  f.write("Python\n")
  f.write("is\n")
  f.write("easy\n")
  f.write("language\n")
  print("is file closed ?",f.closed)
  print("is file closed ?",f.closed)
print("Now is file closed",f.closed)
```

```
is file closed ? False
is file closed ? False
Now is file closed True
```

```
f=open("myfile.txt",'r')
print(f.tell())
print(f.read(2))
print(f.tell())
print(f.read(4))
```

```
0
Py
2
thon
```

```
data = "Learnng python is very easy from Ashish Sir"
with open("myfile.txt", 'w') as f:
    f.write(data)
with open("myfile.txt", 'r+') as f:
    fileData = f.read()
    print(fileData)
    print("Current cursor position:", f.tell())
    f.seek(15)
    print("Current cursor position:", f.tell())
    f.write("-JAVA-")
    f.seek(0)
    fileData = f.read()
    print(fileData)
```

```
Learnng python is very easy from Ashish Sir
Current cursor position: 43
Current cursor position: 15
Learnng python -JAVA-y easy from Ashish Sir
```

```
import os,sys
frame=input("Enter file name:")
if os.path.islife(fname):
  print("File exist:",fname)
  f=open(fname,'r')
else:
  print()
```